Order Statistics & Bivariate Normal Distributions

STA 312: Statistical Theory I

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Lecture Objectives

1. Introduction to Order Statistics
2. Proofs for Order Statistics
3. Functions of Order Statistics
4. Discrete Continuous Bivariate Distributions
5. The Bivariate Normal Distribution
6. Proofs for the Bivariate Normal Distribution

1. Order Statistics: Conceptual Framework

- Let $X_1, X_2, \ldots, X_n$ be a random sample of size n drawn from a continuous population with Probability Density Function (PDF) $f(x)$ and Cumulative Distribution Function (CDF) $F(x)$.
- Initially, these observations are Independent and Identically Distributed (i.i.d.).
- If we sort these observations in ascending order of magnitude, we obtain:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$$

- The ordered variables $X_{(r)}$ are called the **order statistics**.

1. Order Statistics: Formal Definitions

The r-th Order Statistic

The random variable $X_{(r)}$ denotes the r-th smallest value within the sample.

- **First Order Statistic (Minimum):**

$$X_{(1)} = \min(X_1, X_2, \ldots, X_n)$$

- **Extreme Order Statistic (Maximum):**

$$X_{(n)} = \max(X_1, X_2, \ldots, X_n)$$

- Sorting destroys the mutual independence of the underlying observations. Thus, $X_{(1)}, \ldots, X_{(n)}$ are dependent variables.

2. Proof: PDF of the Sample Maximum $X_{(n)}$ (Step 1)

We seek to derive the PDF of the maximum order statistic, $X_{(n)}$. We begin by analyzing its Cumulative Distribution Function (CDF):

$$F_{X_{(n)}}(x) = P(X_{(n)} \leq x)$$

- For the largest value in a sample to be less than or equal to an arbitrary value x, **every single observation** in that sample must simultaneously be less than or equal to x.
- This allows us to reframe the single event as an intersection of events:

$$F_{X_{(n)}}(x) = P(X_1 \leq x \cap X_2 \leq x \cap \cdots \cap X_n \leq x)$$

2. Proof: PDF of the Sample Maximum $X_{(n)}$ (Step 2)

Because the underlying variables $X_1, X_2, \ldots, X_n$ are mutually independent, the joint probability of their intersection factors into the product of individual probabilities:

$$F_{X_{(n)}}(x) = P(X_1 \leq x) \cdot P(X_2 \leq x) \cdots P(X_n \leq x)$$

Since the sample is identically distributed, each marginal probability matches the baseline population CDF, $P(X_i \leq x) = F(x)$. Substituting this gives:

$$F_{X_{(n)}}(x) = \prod_{i=1}^{n} F(x) = [F(x)]^n$$

2. Proof: PDF of the Sample Maximum $X_{(n)}$ (Step 3)

To arrive at the Probability Density Function $f_{X_{(n)}}(x)$, we take the derivative of the verified CDF with respect to x:

$$f_{X_{(n)}}(x) = \frac{d}{dx}\left(F_{X_{(n)}}(x)\right) = \frac{d}{dx}\left([F(x)]^n\right)$$

Applying the Chain Rule of differential calculus:

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} \cdot \frac{d}{dx}[F(x)]$$

By definition, $\frac{d}{dx}F(x) = f(x)$. Thus, we conclude:

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x) \quad \square$$

2. Proof: PDF of the Sample Minimum $X_{(1)}$ (Step 1)

We turn to the lowest order statistic, $X_{(1)} = \min(X_1, \ldots, X_n)$. We define its CDF and apply the complement rule:

$$F_{X_{(1)}}(x) = P(X_{(1)} \leq x) = 1 - P(X_{(1)} > x)$$

- For the sample minimum to be strictly greater than x, **every single individual observation** must be strictly greater than x.
- This converts into the following joint statement:

$$P(X_{(1)} > x) = P(X_1 > x \cap X_2 > x \cap \cdots \cap X_n > x)$$

2. Proof: PDF of the Sample Minimum $X_{(1)}$ (Step 2)

Utilizing the sample property of mutual independence, we factor the joint survival probability into marginal segments:

$$P(X_{(1)} > x) = P(X_1 > x) \cdot P(X_2 > x) \cdots P(X_n > x)$$

Invoking the survival relationship where $P(X_i > x) = 1 - F(x)$:

$$P(X_{(1)} > x) = [1 - F(x)] \cdot [1 - F(x)] \cdots [1 - F(x)] = [1 - F(x)]^n$$

Substituting this product pattern back into our baseline complement framework yields the explicit CDF:

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$$

2. Proof: PDF of the Sample Minimum $X_{(1)}$ (Step 3)

We differentiate the finalized CDF to obtain the sample minimum's PDF:

$$f_{X_{(1)}}(x) = \frac{d}{dx}\left(1 - [1 - F(x)]^n\right) = 0 - \frac{d}{dx}\left([1 - F(x)]^n\right)$$

Applying the Chain Rule of calculus:

$$f_{X_{(1)}}(x) = -\left(n[1 - F(x)]^{n-1} \cdot \frac{d}{dx}[1 - F(x)]\right)$$

$$f_{X_{(1)}}(x) = -n[1 - F(x)]^{n-1} \cdot (0 - f(x))$$

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x) \quad \square$$

2. The General r-th Order Statistic PDF

Theorem

The PDF of the general r-th order statistic $X_{(r)}$ for $r = 1, 2, \ldots, n$ is given by:

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!}[F(x)]^{r-1}[1 - F(x)]^{n-r} f(x)$$

Heuristic Interpretation:

- $(r-1)$ observations fall below x, each with probability $F(x)$.
- $(n-r)$ observations fall above x, each with probability $1 - F(x)$.
- Exactly 1 observation lands in the infinitesimal interval $(x, x + dx)$, represented by $f(x)dx$.
- The multinomial coefficient accounts for all permutations of these groups.

3. Functions of Order Statistics: Definitions

Order statistics are commonly used to derive key sample estimators.

The Sample Range (R)

The range measures sample variability and is defined as the difference between the maximum and minimum values:

$$R = X_{(n)} - X_{(1)}$$

The Sample Median ($\tilde{X}$)

The central tendency indicator when sorting is applied:

$$\tilde{X} = \begin{cases} X_{\left(\frac{n+1}{2}\right)} & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}+1\right)}\right) & \text{if } n \text{ is even} \end{cases}$$

3. Joint PDF of Extreme Order Statistics

To derive distributions of functions like the Range ($R = X_{(n)} - X_{(1)}$), we require the joint distribution of multiple order statistics.

Joint PDF of $X_{(1)}$ and $X_{(n)}$

The joint density of the minimum and maximum order statistics is given by:

$$f_{X_{(1)},X_{(n)}}(x,y) = n(n-1)[F(y) - F(x)]^{n-2} f(x) f(y)$$

defined over the support region $-\infty < x \leq y < \infty$.

We will use this joint density to perform bivariate transformations.

Joint Probability Mass Function (JPMF)

Let X and Y be two discrete random variables defined on the same sample space. Their JPMF is:

$$p(x, y) = P(X = x, Y = y)$$

Fundamental Conditions:

1. $0 \leq p(x, y) \leq 1$ for all valid coordinate pairs (x, y).
2. The total mass sums to unity over the support space:

$$\sum_x \sum_y p(x, y) = 1$$

4. Discrete Marginals and Conditionals

Marginal PMFs: Derived by summing out the unwanted variable:

$$p_X(x) = \sum_y p(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p(x, y)$$

Conditional PMFs: The probability profile of one variable given that the other is fixed at a specific value:

$$p_{Y|X}(y|x) = \frac{p(x, y)}{p_X(x)} \quad (\text{where } p_X(x) > 0)$$

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)} \quad (\text{where } p_Y(y) > 0)$$

Joint Probability Density Function (JPDF)

For continuous random variables X and Y, the JPDF $f(x, y)$ is a surface representing joint density such that:

$$P((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$$

Fundamental Conditions:

1. $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$.
2. The volume under the joint surface is equal to 1:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$$

Marginal JPDFs: Evaluated by integrating across the entire domain of the secondary variable:

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

Conditional JPDFs:

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} \quad (\text{where } f_X(x) > 0)$$

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} \quad (\text{where } f_Y(y) > 0)$$

4. Proof: Validity of Marginal Density Extraction (Step 1)

Theorem: If $f(x, y)$ is a valid joint continuous PDF, then its extracted marginal component $f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$ is a valid univariate PDF.

Proof.

To prove validity, we must verify that the integrated marginal sums to 1 over its remaining single-variable domain:

$$\text{Target Verification: } \int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

Substitute the formal definition of the marginal function into our target verification expression:

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x, y)\, dy \right] dx$$

□

4. Proof: Validity of Marginal Density Extraction (Step 2)

Proof Continued.

By applying Fubini's Theorem for continuous, non-negative functions over real numbers, we can swap the order of integration:

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x,y)\, dy \right] dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy$$

From the baseline probability axioms of joint distributions, the double integral over the entire plane $\mathbb{R}^2$ evaluates to 1:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy = 1$$

Thus, $f_X(x)$ satisfies all properties of a proper univariate distribution. □

5. The Bivariate Normal Distribution (BVN)

Continuous variables X and Y follow a Bivariate Normal distribution, denoted $(X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, if their joint PDF is:

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} Q(x, y)\right]$$

Where the internal quadratic expression $Q(x, y)$ is given by:

$$Q(x, y) = \left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2$$

Parameters: $\mu_1, \mu_2 \in \mathbb{R}$ (means); $\sigma_1, \sigma_2 > 0$ (variances); $-1 \leq \rho \leq 1$ (correlation).

6. Proof: Marginal PDF of X in the BVN (Step 1)

We wish to show that integrating out y from $f(x, y)$ yields a univariate normal distribution $X \sim N(\mu_1, \sigma_1^2)$. We begin by setting up the marginal integral:

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} Q(x, y)\right] dy$$

To streamline the algebra, we define standardized Z-score tracking variables:

$$u = \frac{x - \mu_1}{\sigma_1} \quad \text{and} \quad v = \frac{y - \mu_2}{\sigma_2}$$

Differentiating the substitution for v with respect to y gives:

$$v = \frac{y}{\sigma_2} - \frac{\mu_2}{\sigma_2} \implies dv = \frac{1}{\sigma_2} dy \implies dy = \sigma_2\, dv$$

6. Proof: Marginal PDF of X in the BVN (Step 2)

Substituting our Z-score variables u and v transforms the quadratic expression $Q(x, y)$ into a standard form:

$$Q(u, v) = u^2 - 2\rho uv + v^2$$

We now complete the square with respect to the variable v inside this expression:

$$Q(u, v) = v^2 - 2\rho uv + \rho^2 u^2 + u^2 - \rho^2 u^2$$

Grouping the terms reveals a perfect trinomial square:

$$Q(u, v) = (v - \rho u)^2 + u^2(1 - \rho^2)$$

6. Proof: Marginal PDF of X in the BVN (Step 3)

Next, we substitute this completed-square form back into our exponential kernel:

$$\exp\left[-\frac{1}{2(1-\rho^2)}Q(u,v)\right] = \exp\left[-\frac{(v-\rho u)^2+u^2(1-\rho^2)}{2(1-\rho^2)}\right]$$

Distributing the denominator across both independent numerators gives:

$$= \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)} - \frac{u^2(1-\rho^2)}{2(1-\rho^2)}\right]$$

Using exponential rules ($e^{A+B} = e^A \cdot e^B$), we separate the terms:

$$= \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right] \cdot \exp\left[-\frac{u^2}{2}\right]$$

6. Proof: Marginal PDF of X in the BVN (Step 4)

We substitute these modified terms and our differential $dy = \sigma_2 dv$ back into our full marginal integral expression:

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{u^2}{2}\right] \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right] (\sigma_2\, dv)$$

Canceling out the common standard deviation term σ_2:

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \exp\left[-\frac{u^2}{2}\right] \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right] dv$$

We move all terms that do not depend on the integration variable v outside the integral:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{u^2}{2}\right] \left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right] dv \right)$$

6. Proof: Marginal PDF of X in the BVN (Step 5)

Let us isolate and evaluate the term within the large brackets:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right] dv$$

Observe that this matches the exact probability density function of a standard univariate Normal distribution curve where:

- The integration variable is v.
- The mean parameter is equal to ρu.
- The variance parameter is equal to $(1-\rho^2)$.

Because the definite integral of any valid probability distribution over its full support domain equals 1, this entire bracketed expression simplifies to 1.

6. Proof: Marginal PDF of X in the BVN (Step 6)

Substituting 1 back into our main equation simplifies the marginal density expression to:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{u^2}{2}\right] \cdot 1$$

We now reverse our original substitution by replacing the variable u with $\frac{x-\mu_1}{\sigma_1}$:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{\left(\frac{x-\mu_1}{\sigma_1}\right)^2}{2}\right]$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right]$$

This matches the exact structure of a univariate normal distribution, proving that $X \sim N(\mu_1, \sigma_1^2)$. $\square$

6. Proof: Conditional Distribution $Y|X$ (Step 1)

We now derive the conditional distribution of Y given $X = x$. By definition:

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

From our previous factoring steps, the joint distribution $f(x,y)$ can be expressed as:

$$f(x,y) = \left(\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{u^2}{2}} \right) \cdot \left(\frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)} \right] \right)$$

Notice that the first expression in parentheses is exactly our verified marginal PDF, $f_X(x)$.

6. Proof: Conditional Distribution $Y|X$ (Step 2)

Substituting this product form back into our conditional fraction allows us to cancel out $f_X(x)$ from the numerator and denominator:

$$f_{Y|X}(y|x) = \frac{f_X(x) \cdot \frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right]}{f_X(x)}$$

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right]$$

We now expand the terms inside the exponent back out to their baseline variables, x and y.

6. Proof: Conditional Distribution $Y|X$ (Step 3)

We substitute $v = \frac{y-\mu_2}{\sigma_2}$ and $u = \frac{x-\mu_1}{\sigma_1}$ back into the inner exponent expression:

$$v - \rho u = \frac{y-\mu_2}{\sigma_2} - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)$$

Factoring out the standard deviation term $\frac{1}{\sigma_2}$:

$$v - \rho u = \frac{1}{\sigma_2}\left[(y-\mu_2) - \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1)\right]$$

Grouping the terms systematically with respect to the variable y:

$$v - \rho u = \frac{1}{\sigma_2}\left[y - \left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1)\right)\right]$$

6. Proof: Conditional Distribution $Y|X$ (Step 4)

Squaring this expression yields the final form for our exponent's numerator:

$$(v - \rho u)^2 = \frac{1}{\sigma_2^2}\left[y - \left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)\right)\right]^2$$

Placing this back into our conditional density function gives:

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{\left[y - \left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)\right)\right]^2}{2\sigma_2^2(1-\rho^2)}\right]$$

This matches the pattern of a univariate normal density function, proving that:

$$Y|X = x \sim N\left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1),\ \sigma_2^2(1-\rho^2)\right) \quad \square$$

6. Proof: Independence under Zero Correlation (Step 1)

Theorem

For a Bivariate Normal distribution, the variables X and Y are stochastically independent if and only if their correlation parameter $\rho = 0$.

Proof.

We first prove sufficiency ($\Longleftarrow$). Assume $\rho = 0$. We substitute this value into our joint quadratic expression $Q(x, y)$:

$$Q(x,y)\Big|_{\rho=0} = \left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2(0)\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2$$

The cross-product term simplifies to zero:

$$Q(x,y)\Big|_{\rho=0} = \frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2}$$

□

6. Proof: Independence under Zero Correlation (Step 2)

Proof Continued.

Next, we substitute $\rho = 0$ into the normalization constant factor at the front of the joint PDF:

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-0^2}} = \frac{1}{2\pi\sigma_1\sigma_2} = \frac{1}{\sqrt{2\pi}\sigma_1} \cdot \frac{1}{\sqrt{2\pi}\sigma_2}$$

Recombining our constant factor and our simplified exponent terms into the joint PDF equation yields:

$$f(x,y) = \left(\frac{1}{\sqrt{2\pi}\sigma_1} \cdot \frac{1}{\sqrt{2\pi}\sigma_2}\right) \exp\left[-\frac{1}{2}\left(\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right)\right]$$

□

6. Proof: Independence under Zero Correlation (Step 3)

Proof Concluded.

Using exponent multiplication rules ($e^{A+B} = e^A \cdot e^B$), we split the equation into two independent functions:

$$f(x,y) = \left[\frac{1}{\sqrt{2\pi}\sigma_1} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right)\right] \cdot \left[\frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{(y-\mu_2)^2}{2\sigma_2^2}\right)\right]$$

$$f(x,y) = f_X(x) \cdot f_Y(y)$$

Because the joint PDF factors perfectly into the product of its marginal densities, X and Y are stochastically independent.

To prove necessity ($\implies$), if X and Y are independent, their covariance is zero, which naturally implies $\rho = 0$. $\square$

6. Key Properties of the Bivariate Normal Distribution

To summarize our mathematical findings for the Bivariate Normal distribution:

- **Marginal Distributions:** Both marginals are univariate normal: $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$.
- **Conditional Distributions:** Conditional profiles are always univariate normal.
- **Linear Regression:** The conditional expectation $E(Y|X = x)$ forms a straight line with slope $\rho\frac{\sigma_2}{\sigma_1}$, known as the theoretical regression line.
- **Homoscedasticity:** The conditional variance $\text{Var}(Y|X = x) = \sigma_2^2(1 - \rho^2)$ depends solely on the correlation parameter and remains constant across all values of x.

Questions & Answers

End of Lecture Series

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